Twelve positive unknowns for the consistency of ZFC

A separating root construction, 26 integer unknowns,
and a smaller finite coefficient compiler

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Abstract

We combine the current ProveIt blank-machine Diophantine constructions and give further reductions in three measures. First, an explicitly specified polynomial in **12 positive integer unknowns**, of exact total degree **1369**, has a fully charged **144-operation** evaluator. Its solvability is equivalent, by ordinary mathematics formalizable in ZFC, to a finite ZFC refutation. Its three-equation version has maximum degree 1360 and uses 140 operations. The new step combines four auxiliary witnesses into one weighted integer. A divisibility adjustment to a Pell multiplier makes the reconstructed rational witnesses integers; this is essential to soundness. Second, robust separating weights give **26 unrestricted integer unknowns**, degree 3560 and 199 operations; an independent signed-scale argument gives lower-degree alternatives in **27 integer unknowns**. Third, a refined local alphabet and compressed layout reduce the distinguished coefficient from $B = 2^{5^{18}}$ to $B = 2^{5^{16}}$; the new alphabet has 5263 tiles. The archive contains the complete arithmetic circuits, coefficient recipe, finite verification programs, source provenance, and proof notes.

The comparison is pinned to ProveIt commit `f0ae4e0991ca44a7bb4e5671d42e33d775e48035`. These are reductions in specified measures, not a proof of an absolute minimum, a new universal-pair theorem, or a proof of ZFC consistency. The new Diophantine proofs are not kernel checked. The inherited exact-machine endpoint has Lean and Rocq evidence, which is kept distinct from the new work.

Contents

1 Outcome, comparison, and conventions **3**
1.1 The current repository frontier . . . 3

2 The inherited arithmetic interface **4**
2.1 What is used from the signed theorem . . . 5
2.2 Bounds available before restoring any root . . . 5

3 A polynomial in twelve positive unknowns **6**
3.1 The divisibility adjustment and the separating weights . . . 6
3.2 An explicit formula using only polynomial operations . . . 6
3.3 Separation of all algebraic branches . . . 7
3.4 Rationality and the integrality repair . . . 8
3.5 Soundness of the system and the single equation . . . 9
3.6 Completeness of the strengthened multiplier . . . 9
3.7 Exact degree and arithmetic size . . . 10

4 Twenty-seven unrestricted integers by an odd scale **11**

5 Twenty-six unrestricted integers by robust separating weights **13**

6 A 5263-tile compiler and smaller fixed coefficients **15**
6.1 The finite local language . . . 15
6.2 Why the restricted language still describes a blank halt . . . 16
6.3 The four fixed integers . . . 17
6.4 Population and carry conditions . . . 18
6.5 Convolution fields and synchronization . . . 19
6.6 Five-adic control of the actual packed rank . . . 19
6.7 Size and evidence boundary . . . 20

7 Logical meaning, verification, and reproduction **21**
7.1 The equivalence proved in ZFC . . . 21
7.2 What was checked, and what was imported . . . 21
7.3 Files and reproducible commands . . . 22
7.4 Which form to use . . . 23

1 Outcome, comparison, and conventions

There is no single ordering of the constructions by simplicity. The fewest unknowns, lowest degree, shortest arithmetic evaluator, and smallest fixed coefficients favor different presentations. The main new result is a further reduction in the number of positive unknowns, together with independent improvements for unrestricted integers and coefficient size.

Theorem 1.1 (Fixed-coefficient consistency equation)**.** *Bind the four coefficient ports* $(\beta, \kappa, M_C, M_F)$ *to the complete 5263-tile recipe of Section 6. Let* P_{12} *be the ordinary integer polynomial defined in Section 3. Then*

$$\mathrm{ZFC} \vdash \left(\exists \mathbf{x} \in \mathbb{Z}_{>0}^{12}\ P_{12}(\mathbf{x}) = 0\right) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}\,.$$

Consequently,

$$\boxed{\mathrm{ZFC} \vdash \left(\forall \mathbf{x} \in \mathbb{Z}_{>0}^{12}\ P_{12}(\mathbf{x}) \neq 0\right) \longleftrightarrow \mathrm{Con(ZFC)}.}$$

The same existence equivalence holds for the stated three-equation system and for each of the five 27-integer constructions in Section 4, as well as the 26-integer constructions in Section 5.

Here $\mathrm{Ref}_{\mathrm{ZFC}}$ means existence of a finite syntactic proof of contradiction from an ordinary effective presentation of ZFC, and Con(ZFC) is its negation. Section 7 explains the imported calibration and the formalization boundary.

1.1 The current repository frontier

The supplied collection now contains seven wave-2 packages, numbered 09–15. They supersede eight earlier packages. Its latest audited reconciliation explicitly distinguishes several coincident discoveries and corrects earlier degree, domain, and coefficient-binding descriptions [1]. We compare against that wave-2 tree, not the superseded wave-1 frontier.

Construction	Unknowns	Equations	Degree	Ops
Existing minimum-operation product	$15\mathbb{Z}_{>0}$	1	155	67
Existing ordinary system (package 12)	$15\mathbb{Z}_{>0}$	6	28	60
Existing lowest-degree 15-variable system	$15\mathbb{Z}_{>0}$	6	22	61
Existing weighted root product (09)	$13\mathbb{Z}_{>0}$	1	515	85
Existing shorter root product (09)	$13\mathbb{Z}_{>0}$	1	571	84
Existing quartic lift (09,12)	$32\mathbb{Z}_{>0}$	1	4	—
Existing integer ordinary product (14)	$29\mathbb{Z}$	1	188	121
New separating-root product	$\mathbf{12}\mathbb{Z}_{>\mathbf{0}}$	1	1369	144
New separating-root system	$\mathbf{12}\mathbb{Z}_{>\mathbf{0}}$	3	1360	140
New 12-variable sum of squares	$12\mathbb{Z}_{>0}$	1	2720	147
New robust integer product	$\mathbf{26}\mathbb{Z}$	1	3560	199
New robust integer system	$26\mathbb{Z}$	3	3544	195
New robust integer sum of squares	$26\mathbb{Z}$	1	7088	202
New integer ordinary product	$\mathbf{27}\mathbb{Z}$	1	172	116
New integer ordinary system	$27\mathbb{Z}$	6	46	107
New integer ordinary sum of squares	$27\mathbb{Z}$	1	92	124

The symbol $n\mathbb{Z}_{>0}$ means n strictly positive integer unknowns; $n\mathbb{Z}$ means n unrestricted integer unknowns. For a system, degree is the maximum degree of its residuals after all abbreviations are substituted. The comparison counts every binary addition, subtraction, and multiplication, including multiplication by a fixed numeral. Register reuse, constants, and final comparisons are free. If a system is required to output residuals rather than compare its expressions to constants, the new 12-variable residual evaluator costs 142 operations instead of 140. This distinction is recorded explicitly.

The 12-variable result improves the reviewed explicit positive-variable frontier, but its degree and evaluator are larger than those of the 13-variable products. The 26-integer version further reduces the integer coordinate count, at a substantial degree and operation cost. The 27-integer ordinary product improves the previous 29-integer ordinary product in all three displayed numerical measures. Neither change improves the 67-operation positive-variable evaluator.

Coefficient backend	Local tiles	B	Bits of $B-1$
Existing T4 (11)	37075	$2^{5^{18}}$	3814697265625
New refined T4	5263	$2^{5^{16}}$	152587890625

The new base coefficient has exactly 25 times fewer bits. It is still far too large for a convenient expanded numeral: the base alone takes about 19.1 GB in binary. We therefore specify the four integers by exact terminating finite recipes and bind each arithmetic circuit to the whole tuple. A compact recipe does not make these coefficients numerically small.

Classical nine-variable constructions over the nonnegative integers exist (and translate to positive integers), at much larger general degree bounds, and relation combining is a classical technique [9, 10, 11]. The new result concerns this explicitly bound K215 compiler and the displayed circuit. It is not a claim that twelve is the minimum possible number of unknowns, nor that $(12, 1369)$ is a universal pair for all Diophantine sets.

2 The inherited arithmetic interface

The coefficient contract uses $\beta = B-1$, a transport coefficient κ, a low mask M_C, and a *shifted* high mask M_F. If the native high mask is M_F^{nat}, then $M_F = M_F^{\mathrm{nat}} + B - 1$. In particular

$$\begin{gathered} B \geq 16, \quad 0 < M_C < B-1, \quad 0 < M_F^{\mathrm{nat}} < B-1, \\ M_C \equiv 2 \pmod 4, \quad M_F^{\mathrm{nat}} \equiv 4 \pmod 8, \quad \mathrm{pc}(M_C) + \mathrm{pc}(M_F^{\mathrm{nat}}) = \log_2 B. \end{gathered} \tag{1}$$

Those arithmetic properties accompany the full local-tableau, carry, synchronization, and canonical-witness contract. They alone would not be sufficient to encode a machine.

With the common positive coordinates, put

$$\begin{aligned} q &= \beta J + 1, & C &= q - F - \alpha, & X &= wq, & Y &= sq^3, \\ E &= XY, & k &= \eta + \zeta, & c &= kY + \eta, & a &= E + Y, \\ H &= 4a + 3, & \Delta &= a^2 + H = (a+2)^2 - 1, & & & \\ P &= X + ac, & S &= 1 + \Delta c^2, & L &= EkY, & A &= 1 + L(L+k). \end{aligned} \tag{2}$$

Define the actual packed rank and the two retained factors by

$$R = \big((q-1)(q-F) + \alpha\big)(q^2-1) + (M_C + qM_F)J, \tag{3}$$

$$N_k = k - hE - R, \qquad N_t = (\kappa + w)C + (q-F) - z(q-1). \tag{4}$$

All capitalized expressions are abbreviations for substituted polynomials, not additional unknowns.

The earlier normalized construction uses fifteen positive coordinates

$$J, F, \alpha, z, f, h, i_0, T, s, w, \tau, \eta, \zeta, y, \gamma.$$

Its system can be written

$$\begin{gathered} \tau^2 = A, \qquad (P + H\gamma)^2 = S, \qquad f^2 = 1 + \Delta i_0^2 c^4, \\ Q_0(c(Tf - 1) - Rf^2)^2 - (Q_0 - 1)y^2 = 1, \qquad Q_0 = (\Delta i_0 c^2)^2, \\ N_k = N_t = 1. \end{gathered} \tag{5}$$

The inherited theorem identifies existence of such a tuple with a blank halting run of the literal K215 machine [2, 3, 7].

2.1 What is used from the signed theorem

The soundness argument also permits f, T, τ, y, γ, h to be signed, provided the compiler coordinates $J, F, \alpha, z, s, w, \eta, \zeta$ are positive and the normalized multiplier is nonzero. The direct nine-positive, six-signed theorem is proved in package 11's `reduction/SIGNED_DOMAIN_PROOF.md`, copied here as `upstream/signed_domain_core.md`. It also proves the ordinary-strong version. Its hypothesis is met by the positive multiplier used below. The related 29-integer theorem in package 14 adds oddness and a square scale for a different parametrization; those extra restrictions are not needed for the 12-variable reduction.

For the unit-product version, it is enough initially to have $N_k = N_t = \varepsilon \in \{1, -1\}$, with the four norm equations equal to one. The proof establishes $\varepsilon = 1$ after recovering the Pell indices. In particular it does not assume the index or transport sign before the required witnesses have been restored.

Here are the main logical steps of that imported arithmetic proof. Write $\chi_U(n) + \psi_U(n)\sqrt{U^2 - 1} = (U + \sqrt{U^2 - 1})^n$ for integer $U \geq 2$. The first and main equations classify the relevant roots as $k = 2\psi_{2XY^2+1}(n)$ and $c = \psi_{a+2}(p)$. The index equation bounds n; the main norm and normalized strong divisibility give a larger index m with $pc \mid m$. The two literal auxiliary congruences, modulo f and c, give $R \equiv \pm p \pmod{c}$. Growth bounds force $p = R$. The first-index wrap and the negative sign are then excluded. The lower Pell-ratio estimate and the congruence modulo H give $X = 2^R$. Half-binomial extraction and population inversion recover the two native masks, and the coefficient compiler decodes an actual finite computation. The signed main root is controlled by the separate signed-γ argument: its wrong sign would give $X2^R \equiv -1 \pmod{H}$ in a strict interval too small to permit that congruence. These statements, their hypotheses, and the supporting source proofs are included in `upstream/`.

2.2 Bounds available before restoring any root

Suppose only that N_t is a unit. Since

$$N_t = (\kappa + w)C + 1 - F - (z - 1)(q - 1),$$

one has $C \geq 0$. The boundary $C = 0, F = 2, z = 1$ can yield $N_t = -1$, so strict positivity cannot yet be used. The authentic mask bounds give

$$\begin{gathered} q(q^2 - 1) < R < q^4 - q^3 + q^2 - 1 < q^4 - 2, \\ q \geq 16, \qquad X \geq q, \qquad Y \geq q^3 \geq 4096, \qquad E \geq q^4 > R + 2. \end{gathered} \tag{6}$$

Also $kY < c < k(Y+1) < 2kY$, $c > 1$, and $a + 2 < 2E$. These bounds use no eliminated root. That fact makes the proof in the next section noncircular.

3 A polynomial in twelve positive unknowns

The supplied coordinates are exactly

$$\boxed{J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t \in \mathbb{Z}_{>0}.}$$

The four former coordinates γ, τ, f, y are replaced by one coordinate t. The polynomials in (2)–(4) are retained.

3.1 The divisibility adjustment and the separating weights

Define

$$\begin{aligned} &G = Hic^2, && D = 1 + \Delta G^2, && Q = \Delta(D-1) = (\Delta G)^2, && d = Q - 1, \\ &v = cT, && b = c + RD, && e = Q(Dv^2 + b^2) - 1, && B_0 = 2Qvb. \end{aligned} \tag{7}$$

The four equations whose witnesses will be eliminated are

$$(P + H\gamma)^2 = S, \qquad \tau^2 = A, \qquad f^2 = D, \qquad dy^2 = e - B_0 f. \tag{8}$$

The last equality is equivalent, when $f^2 = D$, to

$$Q(vf - b)^2 - (Q-1)y^2 = 1.$$

This is the original normalized system with $i_0 = Hi$, since $vf - b = c(Tf - 1) - Rf^2$. The factor H in G is the new divisibility adjustment. Its role is the exact identity

$$Q \equiv 0 \pmod{H^2}, \qquad \gcd(H, d) = 1. \tag{9}$$

Completeness of this restriction will be proved below.

Use the positive polynomial weights

$$\begin{aligned} M &= 1 + 8THic^2 + 16REH^2i^2c^3, \\ N &= 1 + 2HM. \end{aligned} \tag{10}$$

The coordinate t is intended to encode

$$t = y + M\tau + N\gamma. \tag{11}$$

In particular $N \equiv 1 \pmod{H}$. Neither weight is a new unknown.

3.2 An explicit formula using only polynomial operations

Set $W = Ht + NP$ and $m = HM$. Define the scalar polynomials

$$\begin{aligned} a_0 &= d(W^2 + m^2A + N^2S) - H^2e, \\ a_1 &= -2dWm, && a_2 = -2dWN, && a_3 = 2dmN, \end{aligned} \tag{12}$$

$$\begin{aligned} f_0 &= a_0^2 + Aa_1^2 + S(a_2^2 + Aa_3^2) - H^4B_0^2D, \\ f_1 &= 2(a_0a_1 + Sa_2a_3), \\ f_2 &= 2(a_0a_2 + Aa_1a_3), \\ f_3 &= 2(a_0a_3 + a_1a_2). \end{aligned} \tag{13}$$

Finally put

$$\begin{aligned} U &= f_0^2 + Af_1^2 - S(f_2^2 + Af_3^2), \\ V &= 2(f_0 f_1 - S f_2 f_3), \\ \mathcal{R} &= U^2 - AV^2. \end{aligned} \tag{14}$$

The requested single equation is

$$\boxed{P_{12} = N_k N_t (1 + \mathcal{R}) - 1 = 0.} \tag{15}$$

An equivalent existence condition is the particularly transparent system

$$\boxed{N_k = 1, \qquad N_t = 1, \qquad \mathcal{R} = 0.} \tag{16}$$

Every formula here is an ordinary polynomial with integer coefficients. The formulas contain no variable exponent, division, congruence test, root operation, or hidden quantified parameter. The symbols used in the proof for eliminated roots do not occur as unknowns in the final equation. The emitted shared circuit computes $-f_1$ and $-f_2$ to avoid unnecessary negation gates, and factors $H^4 B_0^2 D$ as $4(H^2Q)^2(Dv^2)b^2$.

The construction is two successive quadratic norms. In the algebra with formal symbols r, u satisfying $r^2 = A$ and $u^2 = S$, the expression

$$\left[d(W - mr - Nu)^2 - H^2 e\right]^2 - H^4 B_0^2 D$$

reduces to $f_0 + f_1 r + f_2 u + f_3 ru$. Taking its norm first in u and then in r gives (14). This explanation is not part of the evaluator; the scalar equations above define the polynomial directly.

3.3 Separation of all algebraic branches

The first task is to show that an integer root of the eliminated equation does not merely encode a collection of irrational or rational nonintegral witnesses. We prove separation before any Pell root is reconstructed.

Lemma 3.1 (Separating inequalities)**.** *Under the positive input assumptions and the weak bounds* (6)*, every algebraic branch of* (8) *satisfies*

$$|y| < M\sqrt{A}, \qquad S > A, \qquad \frac{N\sqrt{S}}{H} > M\sqrt{A} + |y|.$$

The absolute value is the complex absolute value; the first and last inequalities include nonreal auxiliary branches.

Proof. All of G, D, Q, v, b are positive and $Q > 1$. Since $G > 1$,

$$\sqrt{D} < (a+2)G < 2EG.$$

Put $Z_0 = v\sqrt{D} + b > 1$. For either sign of $f = \pm\sqrt{D}$,

$$|y|^2 \le \frac{QZ_0^2 + 1}{Q - 1} \le 2Z_0^2 + 1 < 4Z_0^2.$$

Thus $|y| < 2(v\sqrt{D} + b)$. The right side is bounded above by

$$4cTEG + 2c + 8RE^2G^2.$$

On the other hand, $\sqrt{A} > L = EkY > Ec/2$. Division by this positive quantity gives

$$\begin{aligned}\frac{|y|}{\sqrt{A}} &< 8TG + \frac{4}{E} + 16RE\frac{G^2}{c} \\ &= 8THic^2 + \frac{4}{E} + 16REH^2i^2c^3 < M,\end{aligned}$$

since $E \geq 16^4 > 4$.

For the second claim, $c > kY$ and $\Delta - E^2 > 2EY$ imply

$$\Delta c^2 > k^2Y^2\Delta > k^2(E^2Y^2 + EY) = A - 1.$$

Here $2EY^3 > EY$, as $Y \geq 1$. Consequently $S > A$. Finally

$$\frac{N\sqrt{S}}{H} > 2M\sqrt{S} > 2M\sqrt{A} > M\sqrt{A} + |y|.$$

□

Lemma 3.2 (Injective weighted encoding)**.** *The map $(\gamma, \tau, f, y) \mapsto y + M\tau + N\gamma$ is injective on the finite algebraic solution set of (8).*

Proof. If two tuples have different γ, the absolute difference of their $N\gamma$ terms is $2N\sqrt{S}/H$. By Lemma 3.1, this strictly exceeds $2M\sqrt{A}+|y_1|+|y_2|$, which bounds the remaining possible cancellation. If γ agrees but τ differs, the $M\tau$ difference has magnitude $2M\sqrt{A} > |y_1| + |y_2|$. If both agree and the encoded value agrees, then y agrees. The equality $dy^2 = e - B_0f$, with $B_0 > 0$, forces f to agree as well. □

3.4 Rationality and the integrality repair

Over an algebraic closure, with multiplicities, the exact factorization is

$$\mathcal{R} = d^8H^{16} \prod_{(\gamma,\tau,f,y)} (t - y - M\tau - N\gamma), \tag{17}$$

where the sixteen branches are those of (8). Indeed, for fixed $r = \tau$ and $u = P + H\gamma$, the inner polynomial first factors over the two values of f and then over the two values of y; each of its four linear factors contributes the indicated power of H, and its leading coefficient in t is d^2H^4. Multiplying over the four choices of r, u gives (17).

Suppose $\mathcal{R} = 0$ at integer inputs satisfying the weak bounds. As d, H are nonzero, one of the algebraic tuples in (17) has the specified integer value t. Choose a finite normal extension of $\mathbb{Q}$ containing all its coordinates. Every automorphism over $\mathbb{Q}$ sends it to another allowed tuple and fixes its encoded value. Lemma 3.2 forces the whole tuple to be fixed. The fixed field theorem therefore makes all four coordinates rational. This argument does not assume that the square roots generate independent extensions.

Rationality is only the first half of the argument. Since

$$\tau^2 = A, \qquad f^2 = D, \qquad (P + H\gamma)^2 = S$$

are monic square equations with integer right sides, $\tau, f, u = P + H\gamma$ are integers. Hence the reduced denominator of γ divides H. If $y = r_0/s_0$ in lowest terms, then $dy^2 = e - B_0 f \in \mathbb{Z}$ implies $s_0^2 \mid d$. Thus the denominators of γ and y are coprime by (9). But

$$N\gamma + y = t - M\tau \in \mathbb{Z}, \qquad \gcd(N, H) = 1.$$

Multiplication by N cannot cancel a denominator dividing H, and the remaining two coprime denominators cannot cancel each other. Both therefore equal one. All four reconstructed witnesses are integers.

This last step is the reason for strengthening the Pell multiplier by H. A bare resultant, or a rationality argument without this denominator separation, would leave a real soundness gap.

3.5 Soundness of the system and the single equation

For (16), the weak bootstrap holds directly. The preceding argument restores (8), and

$$vf - b = c(Tf - 1) - Rf^2.$$

The ordinary strong multiplier

$$i_{\text{ord}} = \Delta H i > 0$$

satisfies

$$(i_{\text{ord}} c^2)^2 = \Delta(f^2 - 1) = Q.$$

The reconstructed tuple therefore satisfies the signed ordinary-strong interface exactly. Its positive ports remain positive; f, τ, y, γ may be signed, as permitted by the direct signed theorem in `upstream/signed_domain_core.md`. The witnesses τ, y cannot be zero: $A > 1$, while $y = 0$ would give $Q(vf - b)^2 = 1$ with $Q > 1$. The inherited rank, extraction, and compiler proof gives a blank K215 halt.

For (15), all three integer factors are units. By (14), V is even and

$$\mathcal{R} \equiv U^2 \pmod 4.$$

Thus $1 + \mathcal{R}$ cannot equal -1, whose residue is 3. It equals 1, so $\mathcal{R} = 0$ and $N_k = N_t = \varepsilon$ for a common sign. The weak bootstrap was explicitly proved for these unit signs; the reconstruction consequently remains valid. Applying the signed parent proof resolves $\varepsilon = 1$ and yields the same blank halt. No sign assumption was silently used in the root elimination.

3.6 Completeness of the strengthened multiplier

Start from a canonical accepting blank tableau and its original first/main Pell witnesses. Its actual rank satisfies $R \equiv 3 \pmod 4$, and $c = \psi_{a+2}(R)$. Put $A_0 = a + 2$. Consider the matrix

$$\mathcal{M} = \begin{pmatrix} A_0 & \Delta \\ 1 & A_0 \end{pmatrix}.$$

Its determinant is $A_0^2 - \Delta = 1$. Therefore its powers in the finite group of invertible matrices modulo Hc^2 are periodic. Choose $m_0 > 0$ to be a sufficiently large common multiple of this period and R. Then $\psi_{A_0}(m_0)$ is a positive multiple of Hc^2. Set

$$i = \frac{\psi_{A_0}(m_0)}{Hc^2}, \qquad f = \chi_{A_0}(m_0), \qquad \rho = \Delta\psi_{A_0}(m_0). \tag{18}$$

All are positive integers and $\rho^2 = Q$.

To construct the auxiliary witness explicitly, use

$$V_* = \frac{\chi_\rho(R)}{\rho}, \qquad y = \psi_\rho(R). \tag{19}$$

Because R is odd, $\chi_x(R)$ is divisible by x as an integer polynomial. Write $R = 2j + 1$ and $\chi_x(R) = xQ_j(x^2)$. The inherited elementary Chebyshev identities are

$$Q_j(0) = (-1)^j R, \qquad Q_j(1 - A_0^2) = (-1)^j \psi_{A_0}(R).$$

Now j is odd. Moreover $\rho^2 \equiv 0 \pmod{c}$ and $\rho^2 \equiv 1 - A_0^2 \pmod{f}$. Consequently

$$V_* \equiv -R \pmod{c}, \qquad V_* \equiv -c \pmod{f}. \tag{20}$$

The quantities

$$o = \frac{V_* + c}{f}, \qquad T = \frac{o + Rf}{c}$$

are positive integers. To verify the second integrality claim, note that $of = V_* + c \equiv -R \pmod{c}$, $f^2 \equiv 1 \pmod{c}$, and $\gcd(f, c) = 1$. Hence $(o + Rf)f \equiv 0 \pmod{c}$, so $c \mid o + Rf$. Substitution gives

$$c(Tf - 1) - Rf^2 = V_*.$$

The Pell identity for (19) gives $QV_*^2 - (Q - 1)y^2 = 1$. All other canonical positive coordinates are unchanged. Thus the stronger system has a positive witness whenever the machine halts. No parity restriction on m_0 or on the new i was needed.

Finally take $t = y + M\tau + N\gamma > 0$ using the canonical positive roots. One factor of (17) vanishes, so both the system and the single equation have a positive solution. Together with soundness and the machine calibration, this proves Theorem 1.1.

3.7 Exact degree and arithmetic size

All supplied unknowns have degree one. The relevant degrees are

Expression	c	E, a, H	Δ	R	A, S	G	D	Q, d	b	M	N
Degree	5	6	12	4	22	17	46	58	50	39	45

The exact total degree of $\mathcal{R}$ is 1360; its degree in t is 16. Here is a uniform certificate, rather than a purely syntactic degree bound.

Scale all twelve unknowns by a formal parameter, keeping the fixed coefficients constant. The leading homogeneous forms satisfy

$$A_{\mathrm{top}} = S_{\mathrm{top}} = E_{\mathrm{top}}^2 c_{\mathrm{top}}^2.$$

The branches of $M\tau$ have leading terms $\pm 16 b_{\mathrm{top}}$ of degree 50. The quadratic equation for γ is

$$H^2\gamma^2 + 2HP\gamma + (P^2 - S) = 0,$$

where the three coefficient degrees are 12, 17, and 16; the cancellation

$$P^2 - S = X^2 + 2Xac - Hc^2 - 1$$

is essential. Its branches have orders 5 and -1 in the scaling parameter. The large branch contributes $-64b_{\mathrm{top}}$ to $N\gamma$ at degree 50; the small branch contributes only degree 44. Each y branch contributes $\pm b_{\mathrm{top}}$ at degree 50, independently of the leading choice of f. Thus the sixteen linear factors in (17) have nonzero leading magnitudes

$$15, 17, 15, 17, 15, 17, 15, 17, 47, 49, 79, 81, 47, 49, 79, 81$$

times b_{top}. Equivalently,

$$\mathcal{R}_{1360} = C_0 d_{\mathrm{top}}^{8} H_{\mathrm{top}}^{16} b_{\mathrm{top}}^{16}, \qquad C_0 = 15^4 17^4 (47 \cdot 49 \cdot 79 \cdot 81)^2 > 0. \tag{21}$$

The degree is $8 \cdot 58 + 16 \cdot 6 + 16 \cdot 50 = 1360$. This may be interpreted as a formal Laurent-series calculation over the field of rational functions in the unscaled coordinates. No numerical approximation to an algebraic root is being used. All leading factors are nonzero for every valid fixed coefficient binding with $\beta > 0$.

The leading forms of N_k, N_t have degrees 7 and 2 and are nonzero, so $\deg P_{12} = 1369$. The sum of the three squared residuals has exact degree 2720. The emitted live circuits have the following ledgers:

Presentation	Multiplications	Add/subtract	Total	Degree
Three free comparisons	95	45	140	1360
Three explicit residuals	95	47	142	1360
Single product	97	47	144	1369
Sum of squares	98	49	147	2720

The degree and coefficient in (21) are additionally checked by a modular univariate specialization. The general leading-form argument establishes uniformity on the actual astronomical coefficient slice.

4 Twenty-seven unrestricted integers by an odd scale

The ordinary and normalized fifteen-coordinate certificates admit a smaller integer-domain conversion than the seven-positive-port conversion in package 14. Require positivity only for

$$J, F, \alpha, z, \eta, \zeta,$$

put $i = 2r + 1$, $s = 2v + 1$, and $w = u^2$, and allow $r, v, u, f, h, T, \tau, y, \gamma$ to be arbitrary integers. Replacing each of the six positive ports by its own copy of

$$\mathrm{Pos}(x, y, z) = 1 + x^2 + y^2 + z(z + 1)$$

gives $6 \cdot 3 + 9 = 27$ unrestricted integer unknowns. The map is positive and onto the positive integers: represent $4n - 3$ by three squares and separate its one odd square from its two even squares.

The new point is the signed scale s. The source formulas still give $q \geq 16$, $C \geq 0$, $q(q^2 - 1) < R < q^4 - 2$, and $|Y| \geq q^3$. The usual unit arguments use squares and remain valid for signed roots. The main norm excludes $X = 0$: then $D_{\mathrm{main}} = ac + \gamma H$ and $\Delta = a^2 + H$ would imply $0 \equiv 1 \pmod{H}$. Hence $X \geq q$.

Suppose $Y = -Z < 0$. Define

$$B = (X + 1)Z - 2, \qquad c_0 = -c = kZ - \eta, \qquad P_0 = 2XZ^2 + 1.$$

Then $k(Z-1) < c_0 < kZ$, $P_0 > B$, and $c_0 > k$. The first and main Pell equations yield

$$k = 2\psi_{P_0}(n), \qquad c_0 = \psi_B(p), \qquad p > n, \qquad 2n = R + \varepsilon + v_0|E|,$$

where $v_0 \geq 0$ and $\varepsilon \in \{-1, 1\}$. Thus $n \geq (R-1)/2 \geq 24$, $c_0 > B(B^2-1)^2$, and $c_0 > 2p, 2R$. The normalized strong-rank argument, or its ordinary counterpart, and the two signed auxiliary congruences give $p = R$. The sign change $c = -c_0$ changes only a sign removed when the first congruence is squared. The bound $|E| > R + 2$ excludes wrapping. If $\varepsilon = -1$, Pell duplication and $\chi_B(2) > P_0$ give $c_0 = \psi_B(2n+1) > Bk > kZ$, a contradiction. Therefore $R = 2n - 1 = 2r_0 + 1$.

The elementary Pell growth estimates now imply

$$Z > \frac{c_0}{k} \geq \frac{1}{2}\left(\frac{(2B-1)^2}{2P_0}\right)^{r_0} > \frac{X^{r_0}}{2} > 2^R.$$

The strict middle inequality is justified by

$$(2B-1)^2 - 2P_0X = 4(2X+1)Z^2 - 20(X+1)Z + 25 - 2X > 0.$$

At $Z = 4$ its value is $46X + 9$, and it increases thereafter; here $X \geq 16$ and $Z \geq 4096$. For the original negative Pell parameter $a + 2 = -B$, the two Pell projections modulo $H = 4a + 3$ remain 2^R and 2^{-R}. Since R is odd and $c = -\psi_B(R)$, the two signs of the main root respectively imply

$$X + 2^R \equiv 0 \pmod{H} \quad \text{or} \quad X2^R \equiv 1 \pmod{H}.$$

They contradict $0 < X + 2^R < |H|$ and $1 < X2^R < XZ < |H|$. Thus $s > 0$. The inherited signed compiler theorem now applies. This exclusion only needs a nonzero strong multiplier; the odd parametrization guarantees that requirement.

Completeness follows from an exact parity identity, rather than an additional padding conjecture. Write $q = 2^t$, $M_0 = q^2$, and

$$S_0 = (C-1) + qF, \qquad T_0 = (M_C J + 1) + q(M_F^{\mathrm{nat}} J - 1),$$

where M_F^{nat} is the unshifted mask. Native mask disjointness gives $S_0 \,\&\, T_0 = 0$, and $\mathrm{popcount}(T_0) = t + 2$. Consequently

$$R = (M_0 - 1 - S_0)M_0 + (S_0 + T_0), \qquad \mathrm{popcount}(R) = 3t + 2.$$

The half-binomial identity gives $v_2(Y) = 3t$, so $s = Y/q^3$ is odd. The existing odd-multiplier and square-w completeness choices are unchanged. The new substitution costs $6 \cdot 7 + 2 + 2 + 1 = 47$ gates.

Construction	Equations	Degree	Gates
Normalized product	1	274	114
Reduced normalized product	1	216	115
Ordinary product	1	172	116
Shifted ordinary system	6	46	107
Ordinary sum of squares	1	92	124

All rows use 27 unrestricted integers. The full signed proof, emitted circuits, and reproducible arithmetic checks accompany the report in `proofs/ODD_SCALE_27_PROOF.md`. The inherited signed and ordinary rank arguments are those of packages 14 and 12 [7, 6].

The degrees are uniform over every fixed $\beta > 0$. To see this without a numerical coefficient specialization, write p_* for the highest homogeneous part of a substituted port, and set

$$Q_* = \beta J_*, \quad k_* = \eta_* + \zeta_*, \quad C_* = Q_* - F_* - \alpha_*, \quad N_{t,*} = w_* C_* - z_* Q_*.$$

Here $w_* = u^2$, $s_* = 2v$, and $i_* = 2r$. Each guarded highest part is a sum of three squares in its own disjoint variables. In particular $N_{t,*} \neq 0$, because its specialization at $u = 0$ is $-z_* Q_* \neq 0$. The highest parts of the three product polynomials in table order are

$$\begin{aligned} -8Q_*^{91} h\gamma i_*^4 k_*^{13} w_*^{13} s_*^{26} T^2 f^2 N_{t,*}, \\ -8Q_*^{71} h\gamma i_*^2 k_*^9 w_*^{11} s_*^{20} T^2 f^4 N_{t,*}, \\ 8Q_*^{55} h\gamma i_*^2 k_*^9 w_*^7 s_*^{16} T^2 f^4 N_{t,*}. \end{aligned}$$

These are nonzero polynomials of degrees 274, 216, and 172. The unique degree-46 system residual has highest part $a_*^2 c_*^2 T^2 f^4$, where $a_* = w_* s_* Q_*^4$ and $c_* = k_* s_* Q_*^3$; its square gives the exact SOS degree 92. No value of the other fixed coefficient ports affects these conclusions.

5 Twenty-six unrestricted integers by robust separating weights

The twelve-coordinate norm also admits an integer-domain refinement. Keep seven positive ports $J, F, \alpha, z, i, \eta, \zeta$, replace them by disjoint copies of Pos, put $s = 2v + 1$ and $w = u^2$, and leave h, T, t, v, u unrestricted. This uses $7 \cdot 3 + 5 = 26$ integers. In this construction i is positive, with no assertion that it is odd. That distinction is necessary because the normalized multiplier is constrained to be divisible by H.

Use the common prefix and write

$$\begin{aligned} P = X + ac, \quad S = 1 + \Delta c^2, \quad L = EkY, \quad A = 1 + L(L + k), \\ G = Hic^2, \quad D = 1 + \Delta G^2, \quad Q = \Delta(D - 1), \quad d = Q - 1, \\ v_0 = cT, \quad b = c + RD, \quad e = Q(Dv_0^2 + b^2) - 1, \quad B_0 = 2Qv_0 b. \end{aligned}$$

Replace the positive-domain separating weights by

$$\begin{aligned} U_0 &= (T^2 + 1)(a^2 + 5)H^2 i c^4, \\ M &= 1 + 2(U_0 + b), \qquad N = 1 + H^2 M(A + 1). \end{aligned}$$

Use these weights in the same explicitly expanded two-quadratic-norm polynomial, denoted $\mathcal{N}_{26}$. The formulations are

$$N_k = 1, \quad N_t = 1, \quad \mathcal{N}_{26} = 0, \qquad \text{or} \qquad N_k N_t (1 + \mathcal{N}_{26}) - 1 = 0.$$

Here are the additional signed-domain details. The unit guard is valid because the inner norm coefficient is even, so $\mathcal{N}_{26}$ is a square modulo four. Weak transport therefore gives $C \geq 0$ and $0 < R < q^4 - 2$ before roots are recovered. Even if $X = 0$, the odd scale ensures $|a| \geq q^3$, $|c| > 1$, and

$$\Delta \geq (q^3 - 1)(q^3 - 3) > q^4 > R, \qquad S > 1, \quad A \geq 1.$$

Also $D > \Delta + |c|$, which gives the crucial interval

$$1 < b < Q, \qquad Q - b = (\Delta - R)D - \Delta - c > 0.$$

Consider the algebraic triples satisfying

$$(P + H\gamma)^2 = S, \qquad \tau^2 = A, \qquad (dy^2 - e)^2 = B_0^2 D.$$

Every complex auxiliary branch obeys $|y| < 2(|v_0|\sqrt{D} + b) < M$. Indeed $\sqrt{D} < |a+2|\,|G|$, and the factors in U_0 bound respectively $|T|$, $|a+2|$, $|H|$, and $|c|^3$ by integer polynomials. Furthermore

$$\frac{N\sqrt{S}}{|H|} > M(A+1) \geq M\sqrt{A} + M > M\sqrt{A} + |y|.$$

Thus $(\gamma, \tau, y) \mapsto y + M\tau + N\gamma$ is injective on distinct triples, including at $A = 1$ and when the auxiliary quartic has repeated roots. The exact norm factorization and Galois invariance of the supplied integer t imply that γ, τ, y are rational.

Both τ and $P + H\gamma$ are integers, so the denominator of γ divides H. For $y = r/s$ in lowest terms, the quartic

$$d^2 y^4 - 2dey^2 + e^2 - B_0^2 D = 0$$

shows that every prime dividing s divides d. Because Q is divisible by H^2, $\gcd(d, H) = 1$. The identity $y + N\gamma = t - M\tau$ therefore makes y integral, and $N \equiv 1 \pmod{H}$ then makes γ integral. This argument does not assume that T or B_0 is nonzero.

If $T = 0$, then $dy^2 = Qb^2 - 1$, hence

$$d(y^2 - b^2) = b^2 - 1 > 0.$$

Integrality gives $|y| \geq b+1$, whereas $b < Q$ gives $d \geq b$. The left side is then at least $b(2b+1) > b^2 - 1$, a contradiction. Therefore $B_0 \neq 0$, and

$$f = \frac{e - dy^2}{B_0}, \qquad f^2 = D$$

is an integer. The full normalized strong and auxiliary equations have now been recovered. The main norm excludes $X = 0$ modulo H. The negative-scale argument of the preceding section applies to the nonzero normalized multiplier Hi and excludes $s < 0$. The inherited signed compiler theorem then proves soundness.

For completeness, retain the canonical square w and odd s. Choose the strong Pell index to be a common multiple of R and a period of the invertible Pell matrix modulo Hc^2. This gives a positive $i = \psi_{a+2}(m)/(Hc^2)$ and the required auxiliary quotient, without requiring odd i. The canonical roots make $t = y + M\tau + N\gamma$ an integer, and all seven positive ports have Pos representations.

The exact integer degrees are

$$\deg d = 106, \quad \deg b = \deg M = 92, \quad \deg N = 154, \quad \deg(P^2 - S) = 29.$$

The two main-root branches have degrees 9 and -2; therefore the $N\gamma$ term dominates every linear factor of the norm. Its leading form is

$$(\mathcal{N}_{26})_{3544} = d_{\mathrm{top}}^8 N_{\mathrm{top}}^{16} (H_{\mathrm{top}} c_{\mathrm{top}}^2)^8 \neq 0.$$

The comparison system has degree 3544 and 195 gates; the product has degree 3560 and 199 gates. A sum of squares has degree 7088 and 202 gates. All counts include the 52 substitution gates. These constructions provide a variable-count improvement at a substantial degree and circuit cost; the 27-integer degree-92 polynomial remains a useful alternative.

6 A 5263-tile compiler and smaller fixed coefficients

This section supplies the complete coefficient tuple used in Theorem 1.1. It combines two independent changes: a smaller finite local language and a more compact placement of its copy fields. The arithmetic equations in the preceding sections do not change. Relative to package 11's T4 backend, the tile count falls from 37075 to 5263 and the fixed base falls from $2^{5^{18}}$ to $2^{5^{16}}$. The general native compiler and its Pell interface are inherited from the established blank-input construction [2, 5, 3]; the new work below verifies the local language and all changed layout hypotheses.

The exact machine table is included as `coefficients/k215.tm`, with its literal 430-action projection in `coefficients/k215_transitions.json`. The ordered new alphabet is `coefficients/windows5263.json`; its generating predicate and finite checks are in `coefficients/compiler5263.py`. These files give a terminating finite definition of the coefficients, without running K215 until it halts. In particular, the definition does not assume that a ZFC refutation exists.

6.1 The finite local language

There are 439 cell symbols. Codes 0 and 1 denote headless tape bits. For $q \in \{0, \dots, 215\}$ and $b \in \{0, 1\}$, code $2(q+1)+b$ denotes bit b carrying a head in state q. Use the five additional symbols

$$L_0 = 434, \quad Q_0 = 435, \quad R_0 = 436, \quad V_0 = 437, \quad H_0 = 438.$$

The first three are initialization phases, all with blank payload; only Q_0 carries a head, in state 0. The symbols V_0, H_0 are the vertical and horizontal boundaries and carry no tape payload. The literal transition entering state 215 retains its original write and movement. Only after entering state 215 do we use the completion $(215, b) \mapsto (b, 0, 215)$ for padding.

A tile is a tuple (l, c, r, n) giving left, center, right, and next-time center cells. The following horizontal pair rule will be used on both (l, c) and (c, r):

1. Two V_0 cells cannot be adjacent. Immediately right of V_0 an initialization phase must be L_0; immediately left of V_0 it must be R_0.
2. If neither member is V_0, the pair agrees on being H_0 and agrees on being an initialization phase. When both are phases, the allowed ordered pairs are $L_0L_0, L_0Q_0, Q_0R_0, R_0R_0$.

Apart from the exceptional forbidden tile specified below, admit exactly the tuples with these properties:

1. At most one of l, c, r carries a head, and both horizontal pair rules hold.
2. If V_0 occurs in l, c, r, every ordinary cell in that triple is the blank headless cell 0, and Q_0 is absent.
3. The center c is V_0 if and only if n is V_0. A center V_0 imposes no further transition condition.
4. If $c = H_0$, then $n \in \{L_0, Q_0, R_0\}$.
5. At an interior center, a head cannot move into an adjacent V_0. If $n = H_0$, every head in the horizontal triple is in state 215.
6. Otherwise, at an interior center, n is an ordinary nonphase cell given by the exact deterministic three-neighbor update. Its bit is the bit written at the center, or the unchanged bit if the center is headless. Its head, if any, is the unique incoming head from the left, center, or right according to the completed transition table. Here V_0 is interpreted as blank and headless, and H_0 cannot supply an interior update.

The retained original one-head restriction makes the successor unambiguous. The blank-margin restriction also excludes a head next to a vertical boundary, so the no-escape clause is redundant for such triples; it is stated to make the relationship to the original T4 predicate explicit.

Put the two distinguished tiles first:

$$S_0 = (434, 435, 436, 1), \qquad E_0 = (438, 0, 0, 438). \tag{22}$$

The tile S_0 passes the displayed rules: K215's first action is $(0, 0) \mapsto (1, +1, 93)$. Admit E_0 as the sole exception to the additional restrictions. It was already a valid original T4 tile, but it has no global extension: a tile immediately to its left would have center H_0 and right neighbor 0. Such a tile is not E_0 and violates the horizontal pair rule. Thus using E_0 as selector 1, which the arithmetic forbids, discards no actual computation.

Order S_0 first, E_0 second, and all remaining passing tuples lexicographically in the displayed coordinate order. The finite rule, the literal transition table, and this ordering determine the alphabet completely.

Proposition 6.1 (Exact alphabet size)**.** *The ordered alphabet has exactly 5263 tiles.*

Proof. For a headless ordinary center and no vertical boundary, the two neighbors give $2^2 + 2 \cdot 432 \cdot 2 = 1732$ ordinary successor tiles. A terminal successor allows only the two halted headed symbols, giving $2^2 + 2 \cdot 2 \cdot 2 = 12$ further tiles. Center 0 additionally has the three boundary neighborhoods $(V_0, 0, 0), (0, 0, V_0), (V_0, 0, V_0)$, each with an ordinary and a terminal successor, and the single exceptional E_0. Thus center 0 contributes 1751 tiles and center 1 contributes 1744.

Each of the 430 ordinary working-head center symbols has four choices of its two headless neighbors and one successor, contributing 1720. Each of the two halted-head center symbols has four ordinary and four terminal cases, contributing 16. The centers L_0, R_0 each have three ordinary and two terminal cases; Q_0 has just S_0. A V_0 center has three admissible cells independently on either side, hence nine cases. An H_0 center has two neighbor choices on either side and three phase successors, hence twelve cases. Summing gives

$$1751 + 1744 + 1720 + 16 + 5 + 5 + 1 + 9 + 12 = 5263.$$

This classification is independent of the exhaustive program. The program separately checks all 5215 permitted horizontal triples against all 439 successors, or 2289385 cases, before adding E_0. □

6.2 Why the restricted language still describes a blank halt

Every new tile, including E_0, is an old T4 tile. Therefore every new marked periodic plane is an old marked periodic plane. The inherited finite-strip argument supplies a blank K215 computation ending in halt. This is the soundness direction; restricting the alphabet cannot create an accepting computation.

For completeness, suppose a finite blank halting computation is given. Choose two untouched blank headless tape cells between every visited head position and each vertical boundary. A canonical strip has a full H_0 row, then an initialization row $L_0^u Q_0 R_0^v$, then the true computation, and an arbitrarily long absorbing halted tail. All new horizontal conditions hold. A row immediately before an H_0 row is halted, so the terminal-neighbor condition also holds.

The inherited helical presentation changes the time coordinate by one when crossing a vertical seam. Within distance two of the seam, cells are blank headless symbols, the appropriate boundary-adjacent phases, or H_0. They therefore satisfy the new restriction even when the two sides represent different times. The symbol V_0 separates them, so their phases need not agree across the seam.

Width and height remain arbitrarily large independent integers, and may still be chosen as powers of five. This presentation has one canonical Start and no E_0. Uniqueness of Start is used for this chosen completeness witness, not assumed for arbitrary putative solutions.

6.3 The four fixed integers

Use a subscript c for the clause radix, to distinguish it from the polynomial A in Section 2. Put

$$A_c = 8192 = 2^{13}, \qquad \rho = 2^{78125}, \qquad B = \rho^{1953125} = 2^{152587890625} = 2^{5^{16}}, \tag{23}$$
$$M_0 = 7021, \qquad H_c = 359389, \qquad T_1 = 738963, \qquad T_2 = 1118101.$$

Write $b_c = 78125$ for the binary radix exponent and $L_c = 1953125$ for the cell length, so $\rho = 2^{b_c}$ and $B = \rho^{L_c}$. For $0 \le u \le 438$, define the copy positions

$$e_l(u) = 6580 + u, \qquad e_c(u) = 6141 + u,$$
$$e_r(u) = 5702 + u, \qquad e_n(u) = 5263 + u. \tag{24}$$

The support $\mathcal{E}$ consists of the selector positions $0, \dots, 5262$, the four copy intervals in (24), the dummy position 7019, and the four anchors

$$M_0, 3M_0, 9M_0, 27M_0 = 7021, 21063, 63189, 189567.$$

Thus $|\mathcal{E}| = 7024$. The possible upper dummy spill is position 7020, still strictly below the first anchor. Set

$$\mu = A_c - 2 + \sum_{j=1}^{2840} A_c^j. \tag{25}$$

For the ith tile $w_i = (l_i, c_i, r_i, n_i)$, define

$$v_i = 1 + A_c^{1+l_i} + A_c^{440+c_i} + A_c^{879+r_i} + A_c^{1318+n_i} + \sum_{t=1}^{4} A_c^{1756+t},$$
$$v_{e_l(u)} = A_c^{1+u}, \qquad v_{e_c(u)} = A_c^{440+u},$$
$$v_{e_r(u)} = A_c^{879+u}, \qquad v_{e_n(u)} = A_c^{1318+u},$$
$$v_{3^{t-1}M_0} = A_c^{1756+t} \quad (1 \le t \le 4), \qquad v_{7019} = 0. \tag{26}$$

These cases have disjoint index sets. Now put

$$D_C = \rho^{878} + \rho^{359828} + \rho^{56168} + \rho^{168504} + \sum_{e \in \mathcal{E}} v_e(\rho^{T_1 - e} + \rho^{T_2 - e}),$$
$$D_R = \rho^{H_c},$$
$$M_F^0 = \mu(\rho^{T_1} + \rho^{T_2}) + \sum_{u=0}^{438} \rho^{6141+u} + \rho^{63189} + \rho^{189567}$$
$$+ \sum_{u=0}^{438} (\rho^{H_c+6580+u} + \rho^{H_c+6141+u}),$$
$$M_C = B - 1 - \sum_{\substack{e \in \mathcal{E} \\ e \ne 1}} \rho^e - 2\rho^{7019}, \qquad M_F^{\mathrm{nat}} = M_F^0 + 4. \tag{27}$$

The four ports used everywhere in this article are exactly

$$\boxed{\beta = B - 1, \qquad \kappa = D_C + BD_R, \qquad M_C, \qquad M_F = M_F^{\mathrm{nat}} + B - 1.} \tag{28}$$

The canonical machine-readable specification is `coefficients/T5263_CONSTANTS.json`. Its labels `K` and `MF_source` mean κ and M_F in this article. No individual coefficient may be substituted from another recipe. All exponents and summation limits in (23)–(28) are fixed integers. They define ordinary integer coefficients, with no variable exponent or extra quantified parameter.

6.4 Population and carry conditions

In the standard compiler notation, the new values are

$$\ell = 13, \quad p_c = 1080, \quad m_c = 7023, \qquad m_c = 2\ell + 11 \cdot 439 + 8 + 2p_c.$$

There is one dummy because $m_c + 1 - 5263 - 4 \cdot 439 - 4 = 1$. The clause mask has population

$$\mathrm{pc}(\mu) = \ell - 1 + 2840 = 2852.$$

The two clause bands, three copy-overlap groups, and two tested anchors are disjoint. The extra bit 2 of M_F^{nat} is disjoint from all of them. Hence

$$\begin{aligned} \mathrm{pc}(M_F^{\mathrm{nat}}) &= 2 \cdot 2852 + 3 \cdot 439 + 2 + 1 = 7024, \\ \mathrm{pc}(M_C) &= 5^{16} - 7024. \end{aligned} \tag{29}$$

For the low mask, omitting selector position 1 restores one otherwise removed bit, while subtracting $2\rho^{7019}$ removes the upper dummy bit. Thus its missing population is exactly 7024. The same disjointness and unused high positions give

$$0 < M_C, M_F^{\mathrm{nat}} < B - 1, \qquad M_C \equiv 2 \pmod 4, \qquad M_F^{\mathrm{nat}} \equiv 4 \pmod 8.$$

This verifies the numerical portion (1) of the compiler contract.

To state the carry bound explicitly, let $\Sigma = \sum_{e \in \mathcal{E}} v_e$ and $\mathcal{M} = (7024+2)(2\Sigma+6)$. Every selector coefficient has nine A_c-monomials counting its constant term, while each copy and anchor coefficient has one. Therefore the inherited sufficient radix exponent can be expressed using the fixed integer

$$U_c = 4(7023 + 3)(18(5263 + 4 \cdot 439 + 4) + 6) + 8.$$

It is

$$\begin{aligned} b_{\min} = \max\{&13(4 \cdot 439 + 5 + 1080) + 3, \\ &13(4 \cdot 439 + 4) + \mathrm{bitlength}(U_c) + 1,\ 4\} \\ = 36936. \end{aligned}$$

Our exponent 78125 exceeds it. An exact integer calculation also gives

$$\mathrm{bitlength}(\mu) = 36921, \qquad \mathrm{bitlength}(4\mathcal{M} + 8) = 22909.$$

Thus $\rho > 2\mu + 4$ and $\rho > 4\mathcal{M} + 8$, with ample room. These are the original sufficient carry bounds, including rotations and the optional upper dummy bit; a comparison only of the displayed scalar exponents would not have established that fact.

6.5 Convolution fields and synchronization

At target $e_c(u)$, the bare shift $878 = 2 \cdot 439$ has exactly the source $e_n(u)$. At the targets $H_c + e_l(u)$ and $H_c + e_c(u)$, the bare shift $H_c + 439$ has sources $e_c(u)$ and $e_r(u)$; multiplication by D_R has sources $e_l(u)$ and $e_c(u)$. The two other bare shifts, $8M_0$ and $24M_0$, have precisely the anchor source/target pairs $(M_0, 9M_0)$ and $(3M_0, 27M_0)$. Exact finite set calculations verify all 1319 targets and exclude every unintended hit there.

Set $E_{\max} = 27M_0 = 189567$. Every copy or anchor target is below $T_1 - E_{\max}$. At each clause band T_j, no bare term contributes and the other clause band cannot contribute. Consequently the same clause tests select either an empty cell or one tile together with its four matching copies and four anchors. The field tests yield exactly

$$w_i(r) = w_{i-1}(c), \qquad w_i(c) = w_{i-1}(l), \qquad w_i(n) = w_{i-h_0}(c).$$

Writing $a_i = w_i(c)$, they imply $w_i = (a_{i+1}, a_i, a_{i-1}, a_{i-h_0})$. Thus cyclic data pull back to the actual plane $a_{-x-h_0 t}$ for arbitrary stride h_0. No coprimality or divisibility assumption on that stride is introduced.

The synchronization check must cover arbitrary bit rotations. Here

$$\begin{aligned} T_2 - T_1 = 379138 = 2E_{\max} + 4 \equiv 2 \pmod 4, \\ T_2 + 2E_{\max} + 2 = 1497237 < 1953125. \end{aligned} \tag{30}$$

The support, including the upper dummy spill, lies in $[0, E_{\max}]$. The two possible clause-band contamination sets are therefore disjoint modulo the cell length. The inherited clean-band argument always has an uncontaminated band available.

For a binary rotation $b_c t + \epsilon$, $0 \le \epsilon < b_c$, the carry bound shows that a nonzero remainder cannot satisfy both odd anchor tests: for $1 \le \epsilon \le b_c - 2$ the rotated digits are even, and for $\epsilon = b_c - 1$ only the single upper dummy spill can be odd. Therefore $\epsilon = 0$. For a radix-aligned rotation, the two tests require a support pair separated by $18M_0$. The only such pair is $(9M_0, 27M_0)$, because all nonanchor support is below M_0 and the other anchor differences are distinct. The circumference is too large to introduce a second pair by wrapping. Hence the radix rotation is zero modulo the cell length. These are precisely the two original synchronization steps, with their hypotheses rechecked for the new support.

6.6 Five-adic control of the actual packed rank

The change of copy spacing also changes the residue used in the dummy-bit adjustment, so it must be recalculated. Since 78125 is a power of five, $\rho \equiv 2 \pmod 5$. By (30), each paired clause contribution cancels modulo five, independently of its coefficient v_e. The four bare exponents in D_C are respectively $2, 0, 0, 0$ modulo four. Hence

$$D_C \equiv D_R \equiv 2 \pmod 5, \qquad 2D_C - D_R \equiv 2 \pmod 5. \tag{31}$$

The former residue 3 is not required; the relevant property is invertibility modulo five.

For clarity, use w_0 for the spatial width, H_0^{time} for the temporal height, and $n_0 = w_0 H_0^{\text{time}}$ for their product, to distinguish these quantities from the polynomial coordinates. Choose all three as powers of five, with $H_0^{\text{time}} \ge 25$ and $n_0 > 25d$, where $d = 5^{16}$. The canonical packed base is $q = B^{n_0}$. Switching on the free low dummy bit in cell j changes the *actual* packed rank by $-\Gamma B^j$, where

$$\Gamma = \rho^{7019}(q^2 - 1)\big[1 + q(D_C + BD_R + B^{w_0})\big]. \tag{32}$$

This formula has the original no-wrap conditions $j+1, j+w_0 < n_0$. Because all relevant exponents are powers of five, $B \equiv q \equiv B^{w_0} \equiv 2 \pmod 5$. The bracket in (32) is therefore

$$1 + 2(D_C + 2D_R + 2) \equiv 2D_C - D_R \equiv 2 \pmod 5.$$

Also $\rho^{7019} \equiv 3$ and $q^2 - 1 \equiv 3$ modulo five, so

$$\Gamma \equiv 3 \pmod 5, \qquad \gcd(\Gamma, dn_0) = 1. \tag{33}$$

The unchanged Boolean subset-adjustment theorem on the cells $\{4j : 0 \le j < n_0/5\} \cup \{1\}$ therefore supplies the required congruence $R \equiv dw_0 \pmod{dn_0}$. Width and height padding gives all size and no-wrap conditions. This is a congruence adjustment to the packed expression (3), not the selection of an independent formal exponent.

The canonical origin test still sees Start at selector 0 and the forbidden tile at selector 1. Every copy and dummy position is positive, so their new placement does not change the unit selector contribution at the origin. Dummy changes preserve all tile choices and overlaps. The original positive-slack argument and carry bounds complete the arithmetic witnesses. The odd exponent $d = 5^{16}$ and the canonical rank congruence $R \equiv 3 \pmod 4$ are preserved; these are the padding and parity properties used in the integer construction of Section 4.

Proposition 6.2 (Complete replacement of the coefficient tuple)**.** *The four integers in* (28) *satisfy the full blank-input compiler contract used in Section 2. In particular, substituting this whole tuple into the new arithmetic circuits preserves their equivalence with a blank K215 halt.*

Proof. The local language has the required marked-tableau soundness and padded-helical completeness. Equations (29)–(30) and the exact target checks verify the numerical, carry, field, and synchronization hypotheses of the inherited native compiler. Equations (32)–(33) verify the actual-rank tuning hypothesis. The original Start-origin and slack arguments are unchanged. The blank-input compiler theorem and the arithmetic equivalences therefore apply to this entire tuple [2, 5, 3]. □

6.7 Size and evidence boundary

The coefficient β has exactly 152587890625 bits. The largest port is κ, with exactly

$$d + 78125H_c + 1 = 180665156251$$

bits, because $D_C < B$ while BD_R has its highest bit at $d + 78125H_c$. Thus even this improved tuple has enormous expanded numerals. The supplied finite recipes are exact; the expansions were not materialized.

The program recomputes the complete alphabet, verifies its inclusion in the original T4 alphabet, checks the impossibility of the forbidden tile through every adjacent candidate, and verifies all carry and support conditions above. It also checks 18 finite control tableaux containing 927 windows, including helical seams. Those controls use specified altered transitions to create finite examples and do not assert a halting run of literal K215. The archive records the finite checks separately from the universal mathematical proof.

The arithmetic, finite combinatorics, recurrence arguments, and coefficient definitions used here are formalizable in ZFC. The general compiler and Pell results remain imported mathematical dependencies. Neither these finite checks nor inspection of the archived endpoint reports constitutes a new end-to-end Lean or Rocq verification of the Diophantine equation.

7 Logical meaning, verification, and reproduction

7.1 The equivalence proved in ZFC

The dependency chain is

$$\text{integer solution} \longleftrightarrow \text{finite accepted blank tableau} \longleftrightarrow \mathrm{Halt}(K215, \text{blank}) \longleftrightarrow \mathrm{Ref}_{\mathrm{ZFC}}.$$

The first equivalence consists of the root reconstruction, the signed-domain arguments, the inherited Pell compiler, and the finite coefficient adapter. Completeness runs through the chain in the opposite direction. All integer exponents in the coefficient recipe denote fixed integers; none is a hidden exponentiation in an unknown. Each displayed final object is consequently an ordinary polynomial, or a finite system of ordinary polynomials, over $\mathbb{Z}$.

The final endpoint is the theorem

```
blank_halts_iff_zfc_refutation :
BlankHalts machine215 ↔ BProv ZFCax_s [] Form.fBot
```

in `KEndpoint215.lean`. It has no consistency hypothesis or remaining machine-specific theorem parameter. The Rocq development gives the corresponding equivalence with its `StandardZFC` theory [8].

For precision, `BProv` means existence of a finite list of background axioms and a finitary first-order derivation from them. The letter B refers to the background theory; it does not impose a numerical bound on proof length. The namespace `BoundedZFCConsistency` in some imported modules does not change this endpoint into a bounded-consistency statement. The axiom predicate includes all instances of Separation and Replacement and a standard selector formulation of Choice. We may define Con(ZFC) using this exact effective presentation. The usual finite proof translations between standard equivalent presentations give the same result for another conventional coding [8, 5].

The machine calibration uses finitary syntax, first-order completeness, constructible-universe arguments, and a well-founded quotient-model construction. These are ordinary mathematical constructions available in ZFC; they do not require assuming Con(ZFC). The new algebraic steps use finite extension fields, automorphisms of a finite normal closure, elementary Pell arithmetic, and finite computation. They too are formalizable in ZFC. Composing those proofs gives Theorem 1.1, and classical negation gives its unsolvability version.

This formulation deliberately separates two claims. We supply an ordinary mathematical proof of a theorem provable in ZFC. We do not supply a new machine-checked derivation in an encoded ZFC proof calculus, or a new end-to-end Lean/Rocq formalization of the Diophantine reductions. The external proof-assistant endpoint by itself is not such a translation. No stronger internalization in PA is asserted here.

If ZFC is consistent, the displayed polynomial has no solution in the standard positive integers, or in the standard integers for its signed variants. Under the usual hypotheses of the second incompleteness theorem, ZFC then cannot prove this lack of solutions: the proved equivalence would turn such a proof into a proof of its own consistency. The construction does not determine whether a solution exists. Nor do we infer from mere consistency that ZFC cannot prove the opposite assertion; that stronger independence conclusion would require an additional argument or hypothesis.

7.2 What was checked, and what was imported

The source snapshot is pinned to the full Git commit in the abstract. Files in `upstream/` preserve the inherited texts and their original historical snapshot notices. `SOURCE_MANIFEST.json` records their

retrieval paths, Git blob identities, and SHA-256 hashes. The literal 430-transition machine table is included and compared with the exported transition list before the finite compiler is regenerated.

The archive also preserves two *inherited* endpoint audit logs. The Lean log reports only `propext`, `Classical.choice`, and `Quot.sound`; the Rocq log reports closure under the global context. We inspected these records and the endpoint source. We did not execute a fresh Lean or Rocq build in preparing this archive. Their limited size does not make them a substitute for the complete upstream developments.

The new executable checks are standard-library Python. They verify:

1. Circuit register discipline, liveness, exact operation counts, consistency with independently evaluated norm formulas, and specialized polynomial degree certificates. Structural leading-form arguments in the proofs establish the uniform exact degrees for the authentic fixed coefficient tuple.

2. The signed substitutions, the negative-parameter Pell projection identities, the population identity used to make s odd, and explicit small negative-scale Pell tuples with both main-root signs excluded.

3. The 26-integer norm by a separate implementation of multiplication and conjugation in $\mathbb{Z}[r,u]/(r^2-A,u^2-S)$. This checks the expanded circuit without copying its scalar norm formulas, including $T=0$, $X=0$, and negative-s cases.

4. Every candidate successor for the refined admissible horizontal triples, the exact 5263-tile enumeration, clause/copy geometry, carry bounds, synchronization anchors, mask populations, and the modular unit needed to tune the actual rank. Eighteen finite control tableaux check 927 seam and boundary windows.

Finite tests support the arithmetic specification and catch implementation errors. They do not establish the universal Pell lemmas, the unbounded tableau theorem, or consistency. In particular, the synthetic exact-root fixtures are tests of the algebraic norm identity; they are not claimed solutions for the K215 coefficients. The Galois reconstruction and signed zero-case arguments are proved in the text and independent audit notes, not inferred from testing a sample of integers.

7.3 Files and reproducible commands

From the extracted archive root, run

```
python3 reproduce.py
latexmk -pdf -interaction=nonstopmode -halt-on-error report.tex
```

The first command needs Python 3.10 or later and its standard library only. It builds the new arithmetic packets and the finite compiler in a temporary directory, compares them with the bundled artifacts, checks the literal machine binding, runs the separate quotient-algebra audit, and writes a verification receipt. It does not expand the enormous fixed coefficients. The second command uses a standard TeX installation to rebuild this PDF. The corresponding `Makefile` provides `verify`, `pdf`, and `clean` targets.

Path	Purpose
`report.tex`, `sections/`	Complete article sources.
`data/polynomial12.json`	Fully substituted 12-positive-variable product.
`data/system12.json`	Its three comparisons.
`data/integer26/`	26-integer product, system, and sum of squares.
`data/integer27/`	Five lower-degree 27-integer alternatives.
`code/`	Deterministic circuit builders and independent norm audit.
`coefficients/`	Literal machine, 5263 tiles, coefficient recipe, finite verifier, and inherited endpoint evidence.
`COEFFICIENT_BINDING.json`	Exact mapping from each circuit's port names to the one complete coefficient tuple.
`proofs/`, `upstream/`	New proof/audit notes and inherited proof dependencies.
`comparison.csv`	Machine-readable comparison of the stated alternatives.
`SOURCE_MANIFEST.json`	Pinned source provenance.
`MANIFEST.sha256`	SHA-256 checksums for the distributed files.

7.4 Which form to use

For the fewest positive unknowns in this reviewed family, use `polynomial12.json`; its three-equation system is slightly shorter to evaluate. For unrestricted integer unknowns, the new 26-integer equation uses the fewest coordinates among the explicitly bound variants presented here. If degree or arithmetic size matters more, the 27-integer ordinary system has degree 46 and 107 operations, while its single ordinary product has degree 172 and 116 operations. The unchanged 15-positive-variable product remains the shortest known evaluator in this repository comparison, at 67 operations. A degree-four target still favors the existing quartic lift. The refined coefficient tuple can be used coherently with any version whose complete blank-compiler interface is proved.

The new reductions therefore add points to a tradeoff table. They do not make one equation uniformly simplest. Further useful work would be to formalize the separating-map and denominator lemmas, reduce the large weights while retaining their signed bounds, or compile the smaller numerical proof calculus directly. Those directions are not treated as results of this report.

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